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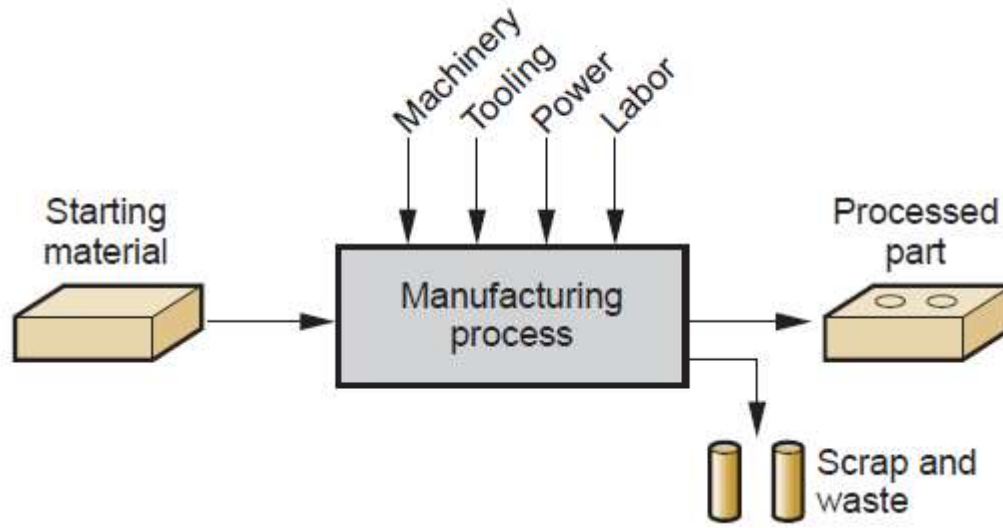
Manufacturing - Making things

- The word *manufacture* is derived from two Latin words,
- *manus* (hand)
- *factus* (make)
- The combination means **made by hand**.

Most modern manufacturing is accomplished by automated and computer-controlled machinery



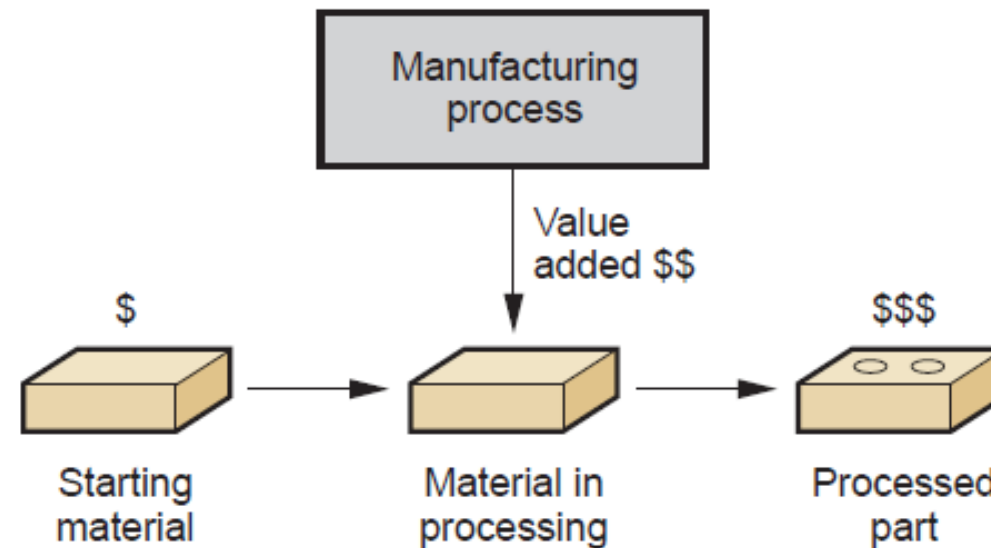
Two ways to define manufacturing



- Technical process

- Economic process

- Iron ore is converted into steel
- Sand is transformed into glass
- Petroleum is refined into plastic
- Plastic is molded into the complex geometry of a chair



Case Study 2 - Industry 4.0 and Smart Manufacturing

- Technology Focus: IoT, Automation, Additive Manufacturing, Data Analytics
- Industry Case: The Transformation of Automotive Manufacturing with IoT
- Analysis: Efficiency, Cost Reduction, and Sustainability Improvements
- Outcome: Evolution of manufacturing processes and predictive maintenance using smart technology.

Industry 4.0 and Smart

Manufacturing: Introduction

- Industry 4.0 and smart manufacturing are the latest technological advancements that have revolutionized the manufacturing industry.
- Industry 4.0 refers to the Fourth Industrial Revolution, which involves the integration of digital technologies such as
 - the Internet of Things
 - Big Data, and
 - Artificial Intelligence
- into the manufacturing process.
- Smart manufacturing is the application of Industry 4.0 technologies to improve the efficiency, productivity, and flexibility of manufacturing systems, processes, and services.
- Industry 4.0 and smart manufacturing have several benefits, such as
 - improved production efficiency
 - reduced costs
 - enhanced product quality, and

Industry 4.0 and Smart Manufacturing:

Introduction

- However, implementing these technologies requires significant investment in infrastructure, skilled personnel, and technology upgrades.
- To address these challenges and fully leverage the potential of Industry 4.0 and smart manufacturing, manufacturers need to develop a clear strategy, which includes identifying the areas of the business that can benefit the most from the technologies.
- In addition, manufacturers need to train their workforce to use and maintain the new technologies, while also addressing concerns about job displacement.
- In summary, Industry 4.0 and smart manufacturing offer significant potential for manufacturers to improve their operations, reduce costs, and stay competitive.
- However, it requires careful planning and investment to fully realize the benefits.

INDUSTRY 4.0: Industrial Revolutions

END OF THE 18TH
CENTURY

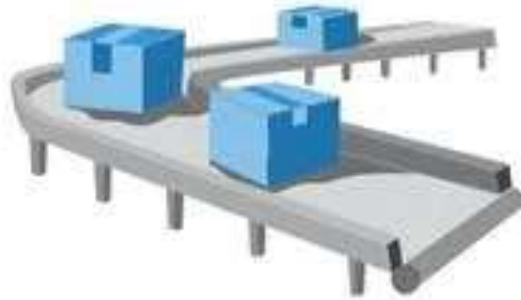


INDUSTRY 1.0 Mechanization

Introduced mechanization of production by using water and steam to increase production capacity and productivity, versus manual craft work

1784 First mechanical loom

START OF THE 20TH
CENTURY



INDUSTRY 2.0 Electrification

Introduced labor-based mass production (assembly lines) powered by electrical energy

1870 First production line, Cincinnati slaughterhouses

START OF THE
1970S



INDUSTRY 3.0 Automatization

Introduced electronics and computers to replace manual work by stand-alone robotic systems

1969 First programmable logic controller (PLC), Modicon 084

PRESENT



INDUSTRY 4.0 Cyber-Physical Systems

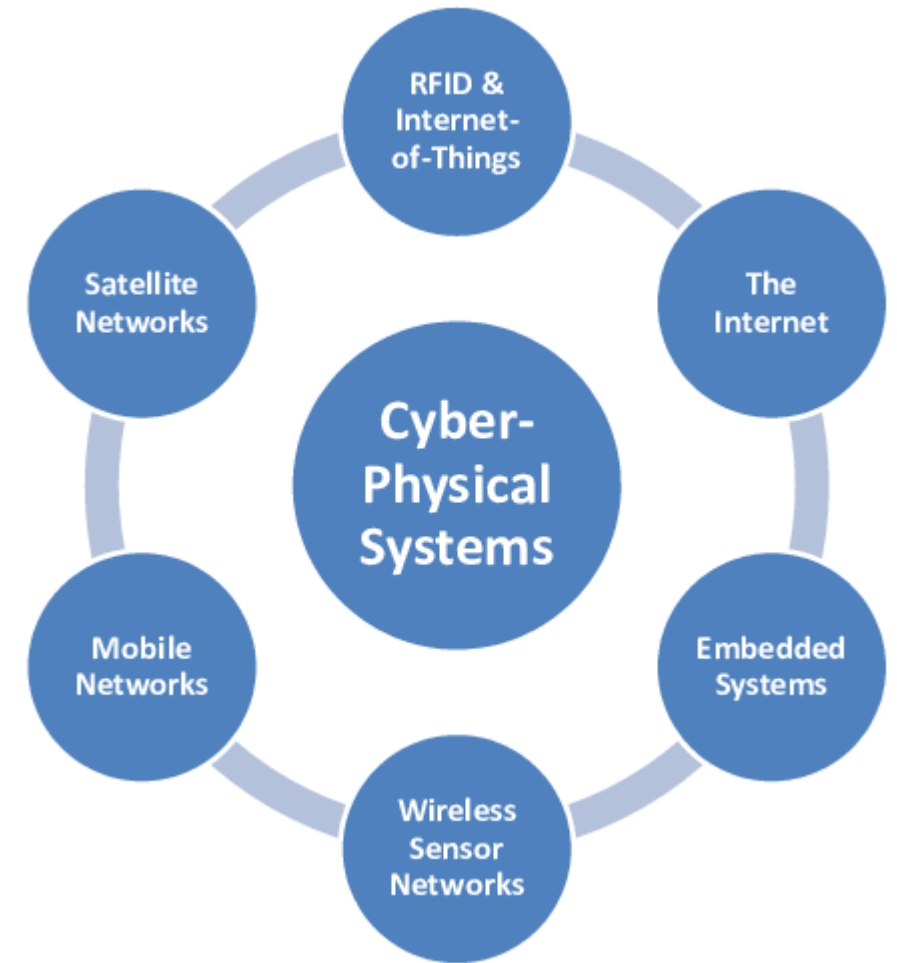
The convergence of physical, digital, and virtual environments through **Cyber-Physical Systems (CPS)** and the **Internet of Things (IoT)**

History: From Industry 1.0 to Industry 4.0 manufacturing

- Industry 4.0 refers to the Fourth Industrial Revolution.
- The word "Revolution" is used because each Industrial Revolution has been powered by some kind of game-changing technology or invention that "revolutionised" the entire industrial world.
- The First Industrial Revolution used **steam**;
- the Second, **electricity**;
- the Third was driven by **basic automation and computing power**
- the Fourth is powered by **cyber-physical systems and intelligent technologies**.
- Today's smart manufacturing is about improving existing factories step by step and enhancing

Cyber-Physical Systems (CPS)

- Cyber-Physical Systems (CPS) are engineered systems that integrate computation, networking, and physical processes.
- They involve close interaction between physical components and computational algorithms, often with real-time, guaranteed performance.
- Essentially, CPS are physical and engineered systems whose operations are monitored, coordinated, controlled, and integrated by a computing and communication core.
- Cyber-Physical Systems (CPSs) are based on the seamless integration of computer algorithms and physical components.
- These systems connect digital and



Industrial Revolutions :First Industrial Revolution

- The first country in the world to begin industrialization is the United Kingdom, which began the development of the cotton industry from a worker based cottage industry into the machine-based economy in the first half of the 18th century.
- With this beginning to the new manufacturing process, the period between about 1760 to 1820 is known for the first industrial revolution in Europe and America.
- In this period, textile industries were the large industries in terms of capital investment, output and the number of employments offered and also the first industry to use modern production technologies

Industrial Revolutions : First Industrial Revolution

- The discovery of the spinning jenny (which produced multiple spools of threads at once) in 1764 by James Hargreaves boosted the generation of machining and invention of coal, the iron, railroads and textiles were the key developments in the first industrial revolution.
- It is time when mechanization commenced, a process that replaced agriculture with industry as the foundations of the economic structure of society.
- Mass extraction of coal along with the invention of the steam engine created a new type of energy that thrust forward all

Industrial Revolutions :Second

Industrial Revolution

- The time between 1830 to 1914 is the period of the second industrial revolution and is also known as technological revolution.
- New technological advancements initiated the emergence of a new source of energy: electricity, gas and oil which give rise to the different inventions.
- As a result, the development of the combustion engine set out to use these new resources to their full potential.
- Furthermore, the steel industry began to develop and grow alongside the exponential demands for steel.
- Chemical synthesis also developed to bring us synthetic fabric, dyes and fertilizer.
- Methods of communication were also revolutionized with the invention of the telegraph and the telephone and so were transportation methods with the emergence of the automobile and the plane at the beginning of the 20th century

Industrial Revolutions :Second

Industrial Revolution

- Chemistry also began its road toward the supply of new artificial materials like Disinfectants and antiseptics, particularly phenol and bromines, role of microbes in the infection of wounds, salicylic acid etc.
- Electric generators, vacuum pumps, gas lighting systems, transformer are the electric systems developed during this revolution.
- Electricity was recognized to be a general system of energy transmission.
- Railways have comparatively become faster, diesel engines, clipper ships were developed in the field of automobiles.
- Steel implements, drainage and irrigation pipes, steam-operated threshers, seed drills, and mechanical reapers were some contributions in the field of agriculture.

Industrial Revolutions : Third Industrial Revolution

- The world economy and the development were also affected by the world war I and II and the industrial developments were again resumed after the second world war and the time period after 1969 is called as the third industrial revolution.
- The electromechanical systems were upgraded into computer-based control systems, where Programmable Logic Controllers (PLC) and industrial robots were the great inventions and implementation in the industrial automation system
- The third industrial revolution appeared with the emergence of a new type of energy whose potential surpassed its predecessors i.e. Nuclear energy.
- This revolution witnessed the growth of the transistor and micro- processors but also the rise of telecommunications and computers.
- This new technology led to the production of miniaturized material which would open doors, most notably to space research and biotechnology.

Industrial Revolutions :Fourth

Industrial Revolution

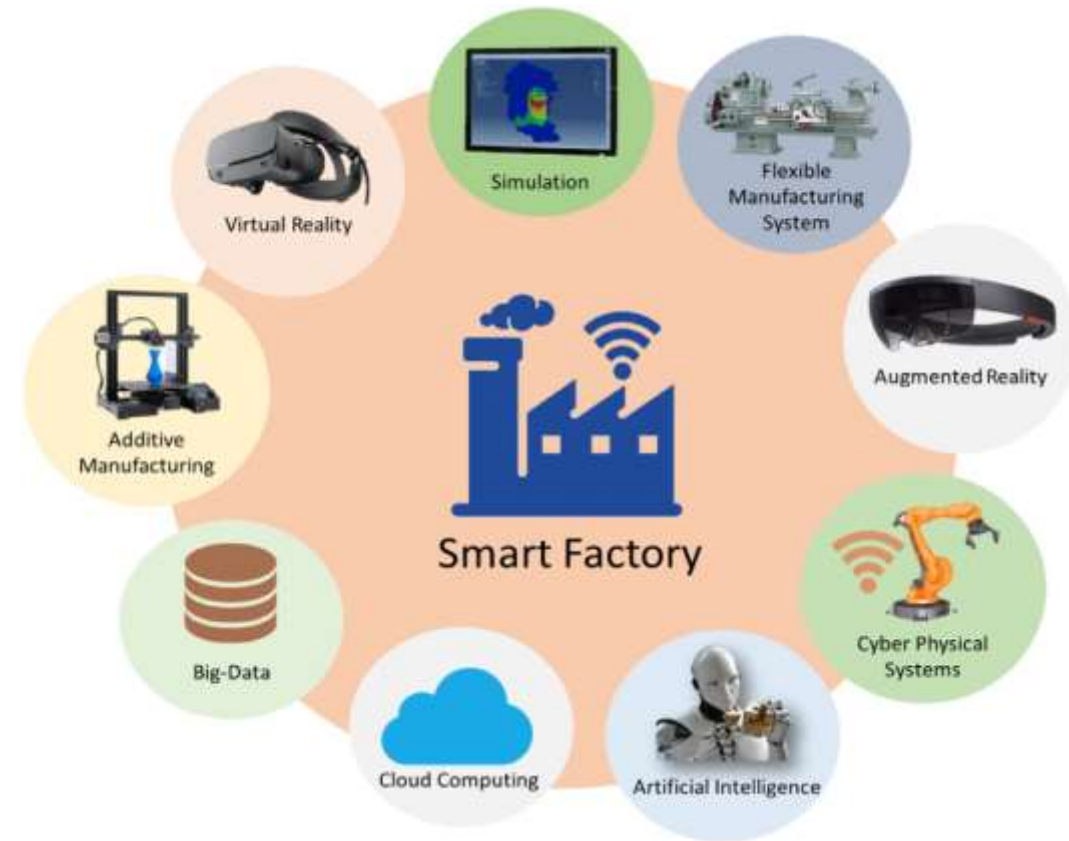
- In 2011, a group of business politics and academic experts introduced the term Industry 4.0 as the fourth industrial revolution to enhance the German competitiveness in the manufacturing industry
- which focuses heavily on interconnectivity through IoT, machine learning and focuses more on the processing of real-time data.
- Industry 4.0 connects internet of things and industrial internet to the manufacturing system to interact with the machines to share their information and make intelligent decisions based on the system algorithm.
- Industry 4.0 basically includes artificial intelligence, automated robots, flexible manufacturing automation systems, additive manufacturing and augmented reality
- Migrating to I4.0 is a gradual process and takes time to upgrade everything from the existing system.

Overview of the System

- The interest of process automation evolves from minimizing human interference in the systems to avoid health hazards and to increase productivity.
- Automation in the manufacturing industry has evolved from the use of basic hydraulic and pneumatic systems to today's modern robots
- The benefits of automation include increased productivity and quality, improved accuracy, saving material costs and energy
- Going through the technological transformations from the past, the wireless-based automation system is being adopted in the different types of system and the use of IoT in them is to ease their operation.
- The use of internet in the control systems is being introduced for the added benefits of remote sensing, distance control, easier data acquisition systems and flexibility of the automation system adopted in the industries as well as in power and energy systems
- IoT is being a topic with a wide scope and emerging technologies, it can be implemented in various

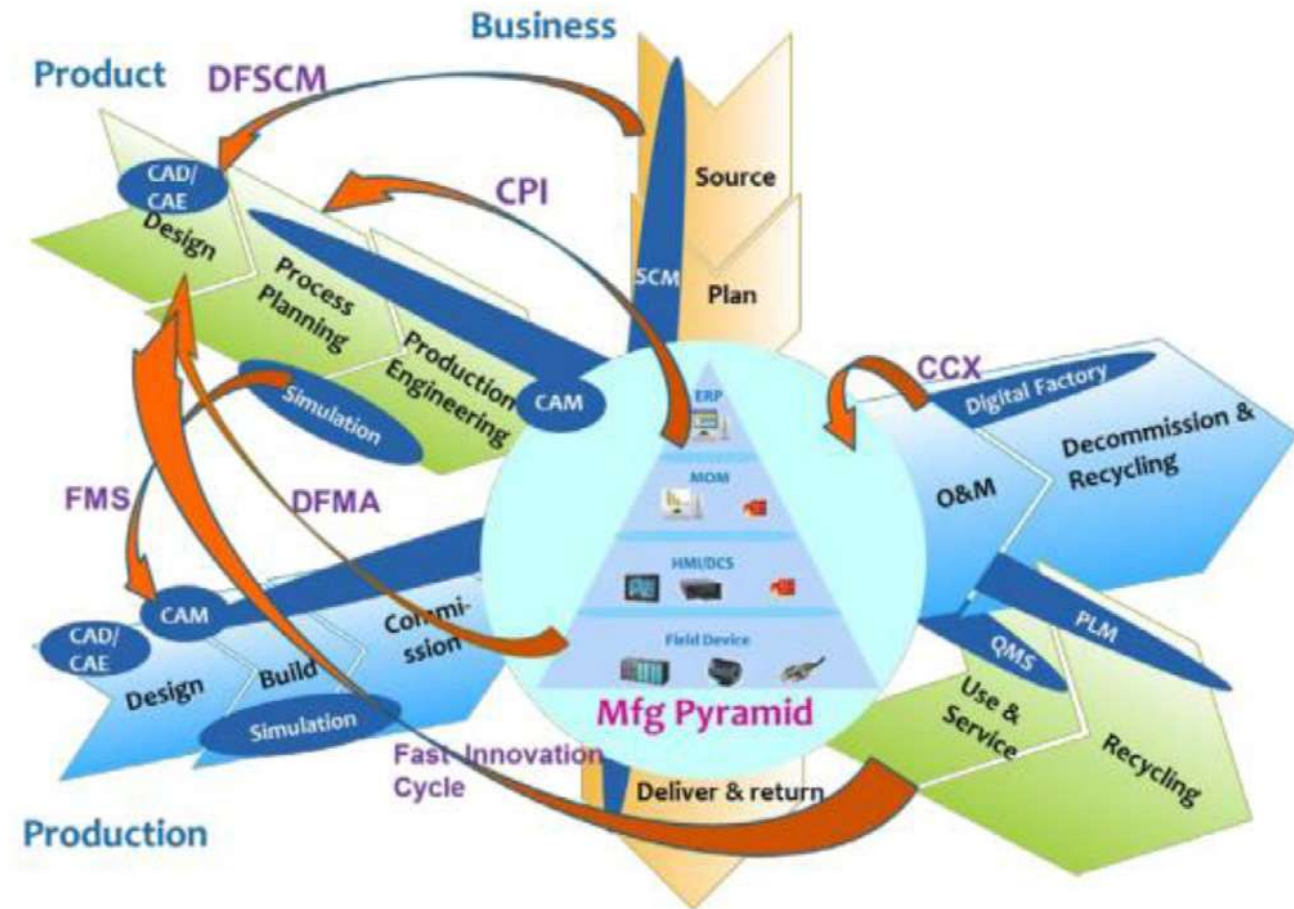
Smart Manufacturing System (SMS)

- Digitization of every part of the manufacturing system with interoperability, real-time control and monitoring, flexible manufacturing, quick response to the market changes, advanced sensors and big data analytics with enhanced productivity
- The SMS operates in two modes;
- semi-autonomous and fully autonomous.
- In the semi-autonomous system, the production engineer defines the goals and sets the parameters in the production system.
- In a fully automatic system, the



NIST's smart manufacturing eco-system model

- **Fig. 4** shows the smart manufacturing ecosystem presented by the National Institute of Standards and Technology (NIST), which describes the interrelation of the related domains of smart manufacturing and their functions.
- This eco-system shows the correlation between the product (green arrow), production process (blue arrow) and business (orange arrow) of the enterprise and



What is smart manufacturing?

- **Smart Manufacturing: A Revolution in Industry 4.0**
- Smart manufacturing, an integral component of the Industry 4.0 revolution, represents a paradigm shift in how goods are produced, bringing together cutting-edge technologies to create highly efficient, interconnected, and adaptive manufacturing ecosystems.
- When we talk about smart manufacturing, we're really talking about data.
- How?
- Smart manufacturing uses real-time data and data-driven technologies like AI and the IIoT -to automatically adapt to changes in customer demand and business needs.
- It uses data from machines and sensors to optimise production, improve quality, and keep equipment running smoothly. And

What is smart manufacturing?

- The Smart Manufacturing Leadership Consortium (SMLC) defines smart manufacturing as “the ability to solve existing and future problems via an open infrastructure that allows solutions to be implemented at the speed of business while creating advantaged value.”
- “Smart manufacturing is becoming the focus of manufacturing globally,” according to an [Infiniti Research](#) expert. “As it permeates the manufacturing process in the form of [smart factories](#) and the adoption of [Industry 4.0](#), it is transforming traditional manufacturing for the better.”
- The rapid evolution of technology is leading this new industrial revolution. According to [MIT Professional Education](#) “This revolution, based on cyber physical production systems, challenges the traditional methods of completing operations in the manufacturing sector, making it increasingly dynamic.”
- “Smart manufacturing consists of the convergence between techniques used in modern data science and artificial intelligence to create processes that can be used in the factory of the future. But why is it necessary today?”
- “Smart manufacturing technology increases efficiency and eliminates points of weakness in the system.
- It’s characterized by a highly connected, knowledge-enabled industrial enterprise where all organizations and operating systems are linked, leading to enhanced productivity, sustainability, and economic performance.”
- Smart manufacturing also makes it possible for manufacturers to use cloud technology to store and use important amounts of data.

Definition of smart manufacturing

- The NIST (National Institute of Standards and Technology) defines
- smart manufacturing as: " fully-integrated, collaborative manufacturing systems that respond in real time to meet changing demands and conditions in the factory, in the supply network, and in customer needs".
- Smart manufacturing is grounded in cloud connectivity.
- It is a combination of human creativity, digitally connected machines and assets, and AI-powered systems and analytics.
- The integration of AI and smart tools helps fuel adaptability and speeds up the capacity to customise outputs based upon real-time data and intel.



- Smart manufacturing refers to the use of **interconnected digital technologies**, both physical and virtual, to digitise and optimise production processes.
- A variety of Industry 4.0 technologies are used for this, resulting in what has been dubbed the 'smart factory'.
- Smart manufacturing relies on the collection of data, the implementation of automation, and the connection of all these technologies to form a symbiotic whole that results in greater efficiencies.

Technologies associated with Smart Manufacturing

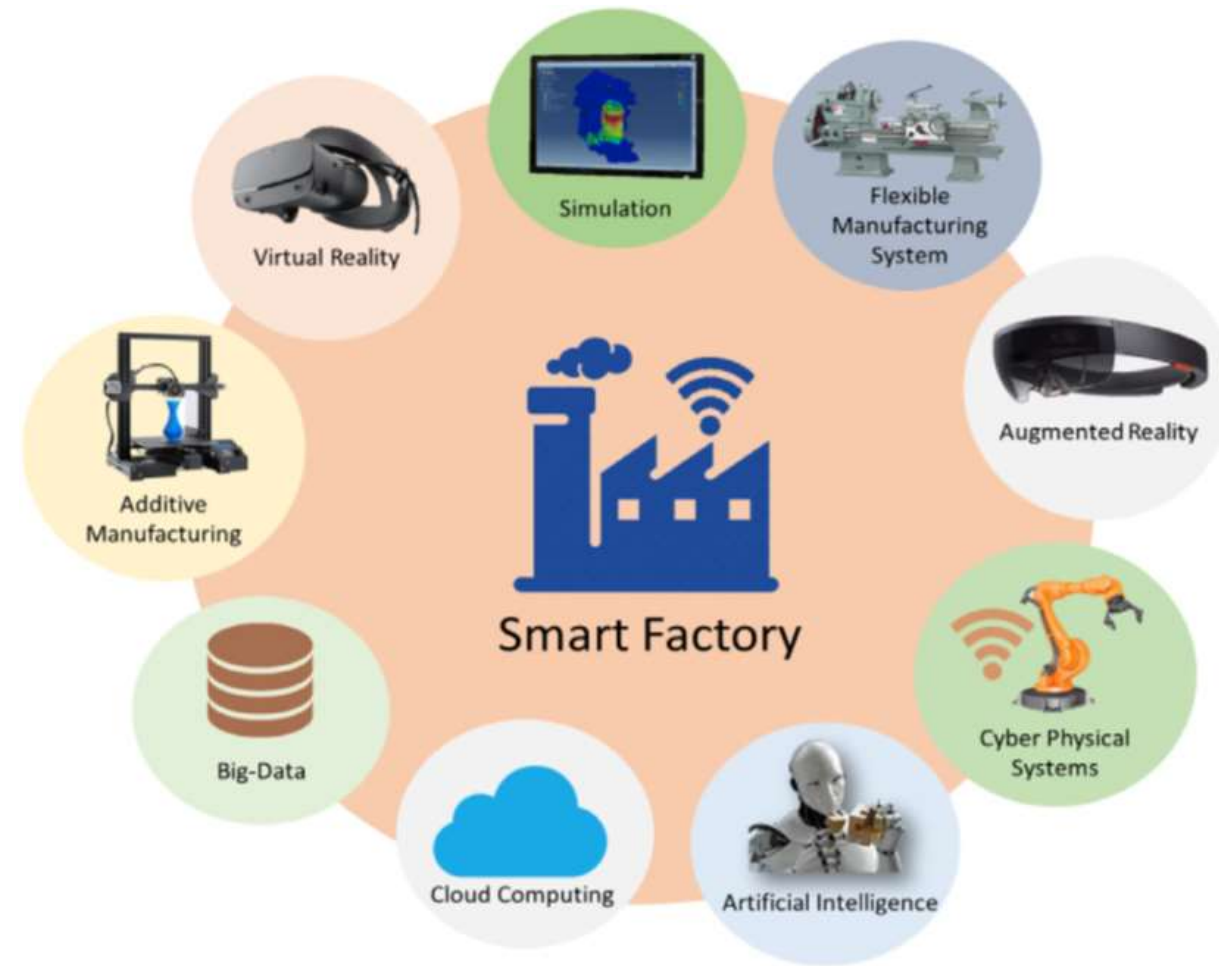
- **1. Simulation (1960s-present)**
- **Origin:** Began with numerical simulation and Monte Carlo methods.
- **Industry Use:** Widely used in engineering and manufacturing for process optimization, modeling, etc.
- **2. Flexible and Reconfigurable Manufacturing Systems (1970s-1980s)**
- **Flexible Manufacturing Systems (FMS):** Emerged in the 1970s with CNC and automation.
- **Reconfigurable Systems:** Developed in the 1990s to allow modular, scalable production systems.
- **3. Virtual Reality (VR) (1980s-1990s; commercial boom ~2010s)**
- Early concepts and headsets in the 1980s;
- Gained mainstream attention in gaming and training by 2010s.
- **4. Augmented Reality (AR) (1990s-2000s: practical use**

Technologies associated with Smart Manufacturing

- **5. Additive Manufacturing (3D Printing) (1980s-2000s; rapid growth after 2010)**
 - First patented in 1984 (stereolithography);
 - Became viable for prototyping and production from 2000s onward.
- **6. Cyber-Physical Systems (CPS) (2006-present)**
 - Term coined in mid-2000s;
 - Backbone of Industry 4.0 - integration of computation, networking, and physical processes.
- **7. Internet of Things (IoT) and Industrial IoT (IIoT) (2010s-present)**
 - IoT conceptualized in late 1990s;
 - Gained momentum with cloud computing and low-cost sensors in the 2010s.
 - IIoT focused on industrial applications like smart factories.
- **8. Big Data Analytics (2010s-present)**
 - Emerged due to the exponential growth of data from IoT, social media, sensors, etc.
 - Used for decision-making, predictive analytics, and quality control.
- **9. Artificial Intelligence (AI) (1950s concept, but Industry 4.0 relevance post-2010)**
 - Classical AI since the 1950s;

Technologies associated with Smart Manufacturing

- Smart manufacturing integrates various technologies related to manufacturing, computing, virtualization, connectivity, data handling etc.
- The scope of smart manufacturing technologies has become broader due to the inter-operation of various technologies resulting in cost-effectiveness, time-saving, easy configuration, better understanding, quick



Technologies and their roles in smart manufacturing

• 1.Virtual Reality (VR)

- provides the facility for experiencing the computer-generated images and videos simulating into real-world activities.
- VR is a wearable device containing video device, audio device, positioning systems such as GPS, external connectivity to other devices and hardware to make the user experience his physical presence in the virtual environment created in simulation.
- VR has been used in the manufacturing systems for training the young engineers and technical graduates who are not well-prepared to deal with the industrial processes are introduced to the manufacturing process, mechanization processes, troubleshooting and maintenance systems through the application of VR which have been more beneficial than the theoretical learning.
- It also helps to reduce the prototyping and testing cost of any product by visualizing the design without making it physically.
- Also, it helps in the digital manufacturing process to visualize and test the products in a simulated environment for the end customers, which expands more options for product customization, renovation and rapid testing of product design.



Technologies and their roles in smart manufacturing

• 2. Augmented Reality

- Artificial environment created across the real-world using computer simulation and can be realized through wearables and mobile devices.
- They use the technology of combining the real physical environment along with the computer-generated graphics to visualize artificially added components to the existing real-world scenario for training, simulation or validation purpose of some of the manufacturing designs before going for real manufacturing.
- This integration of simulated computer graphics to real-world scenarios helps to realize the product in an existing environment. Training to new employees and product testing



Technologies and their roles in smart manufacturing

BMW applies VR and AR technology to production processes

- The BMW Group production team is increasing its focus on effective virtual and augmented reality applications.
- OEM's production academy has been using AR goggles in training sessions for engine assembly units. Visualizations guide participants through all process steps and give specific information.
- Three people can go through the AR training at the same time, after receiving quick guidelines from a trainer who supervises their progress.
- Previously, a trainer had to work with one person at a time, while with the new system this number has increased to three.
- Surveys among participants and



technologies and their roles in smart manufacturing

BMW applies VR and AR technology to production

- Checking a complex part weighing up to 25 tons, such as a **press tool** for the production of body parts, can take a lot of time.
- AR technology, staff at the BMW Group Toolmaking and Plant Engineering unit site in Munich can mount a standard tablet on a tripod, which uses the built-in camera to produce an image of the tool.
- Then, an AR application superimposes this image with the CAD construction data of the tool ordered.
- Based on an average of 50 criteria, such as drill holes and other clear surface features, the staff can see whether all production specifications have been implemented.



Technologies and their roles in smart manufacturing

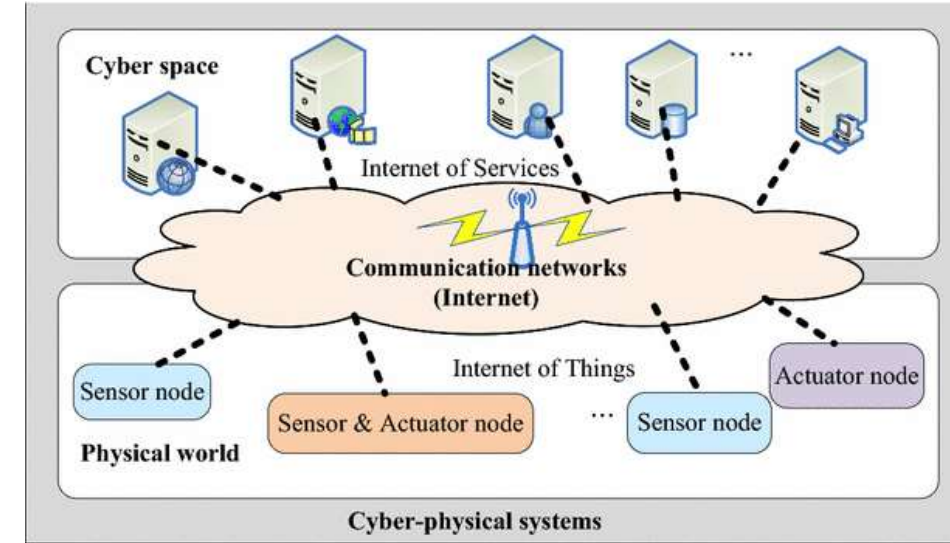
- The **BMW Group Production** focuses increasingly on trendsetting, easy-to-use and effective virtual (VR) and augmented (AR) reality applications.
- VR images, or artificially created images, are ever more realistic and hard to distinguish from real pictures.
- In AR applications, illustrations complement real images.
- AR and VR images can be viewed in special headsets or on normal tablet computers.
- In production, these images are powerful tools in numerous use cases in training and qualification, planning of workstations at the assembly line, or quality control.
- In all applications, the technology keeps modestly in the background.
- No extensive IT expertise is required to use these applications efficiently.
- The car manufacturer BMW uses the XR engineering software Hololight Space to speed up processes by as much as twelve months, from individual vehicle sections through to complex production stages.



Technologies and their roles in smart manufacturing

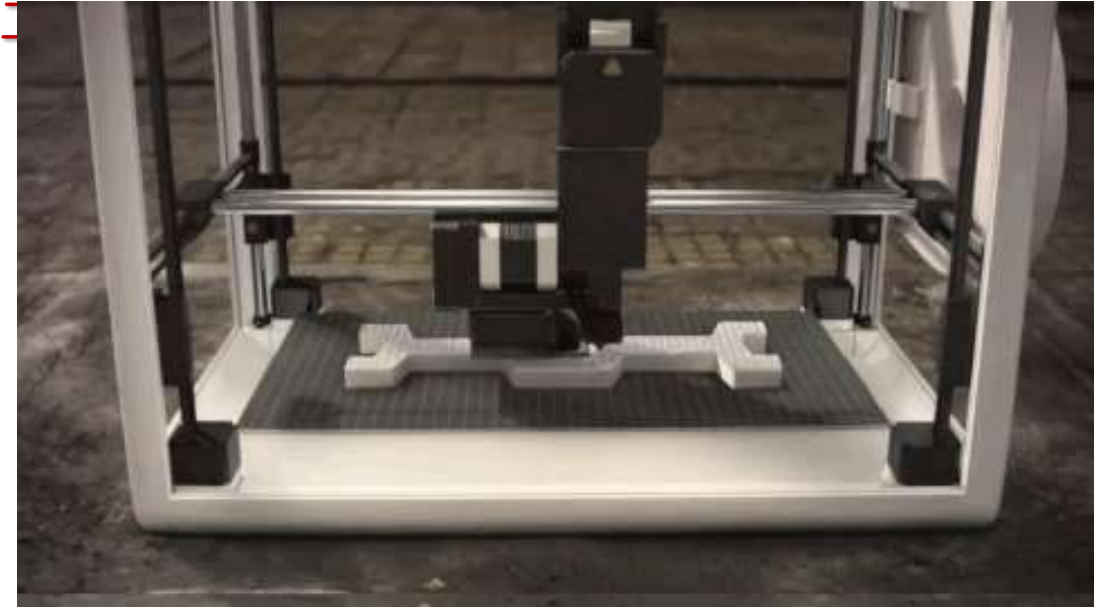
• 3. Cyber-Physical Systems (CPS):

- The industrial processes can be directly monitored and controlled through the use of the internet by the industrial manufacturing engineers, which enables the control engineer to access the control system of the industry from anywhere through cloud computing.
- Integrate cyber technology into the physical processes
- Cyber-Physical System (CPS) collaborates the computational entities to the physical world and its ongoing processes utilizing data processing services available directly on the internet.
- The cloud-based SCADA system can also be defined as the CPS where the sensors, actuators and hardware devices fall under the physical devices and the software, communication system and information exchange through the



Technologies and their roles in smart manufacturing

- 4. Additive Manufacturing:
- The Additive Manufacturing (AM) also known as 3D printing technology creates new horizons towards smart manufacturing technology.
- AM technology is flexible for customization, rapid prototyping, making spare parts quickly and on-site manufacturing which also saves huge time of production and cost in machine tools replacements and raw materials.
- AM also facilitates reverse engineering of any parts or product through 3D scanning and allows reconfiguration in design and quick reproduction for testing and validation is the major contribution of AM in smart manufacturing.
- AM technology is also being widely used in medical science for implants in dentistry and orthopedics for the replacement of injured body parts.
- In civil and architectural



Technologies and their roles in smart manufacturing

• 5. Big Data Analytics

- The collection and analysis of data from various sources like production unit, enterprise, customer feedback and product request system, etc. help in real-time decision making for smart manufacturing.
- Today's manufacturers want their customers to share their feedback and personal view in the products they are using or intend to use, and from that information, manufacturers intend to focus their product design to address the wide range of customers.
- Big data analysis will benefit the manufacturer in the identification of current state and causes of product failures in real-time, driving the customers to buy their products

Operational excellence support

People – Process – Technology
Operational excellence platform

Fall short on any pillar and your OpEx platform becomes tippy

Fall short on two or more pillars and your OpEx platform becomes totally unstable



What is Big Data analytics?

- As data refers to the stored information and recorded operations which were performed by the computer in various forms, big data can be understood as the same data being present in huge amounts and numbers.
- This consists of information provided and both humans and the devices and on almost everything.
- Big data analytics is the collection of the data which is also growing exponentially with time.
- Big data is so large and complex in its size and storage that it is unable to be deciphered and understood efficiently by the traditional data management tools.
- There is a need for much more efficient tools and insights to be able to draw concrete and effective solutions from the available big data.

How do Manufacturers Generate Big Data?

- Big Data is being a very trendy analytical tool these days and is being adopted by manufacturing industries to solve a lot of problems, various kinds of manufacturing industries including aerospace and defense companies, auto manufacturers, heavy equipment manufacturers, electronics companies, oil and gas companies, and other organizations that produce consumer and capital goods are adopting these advanced analytics to gain the benefits most suited for them.
- Apart from the traditional source of big data like conducting online marketing analysis, regulated loyalty programs and social media monitoring the industries can use other software machines to collect such data.
- They use a variety of manufacturing softwares within the company, like MES, CRP, CMMS etc. and

10 Applications of Big Data in Manufacturing

- **1. Production Optimization**

- If that data can generate successful conclusions which will help the manufacturers optimize the solutions and production based on what's most suited for them.

- **2. Maintenance Regulation**

- Big Data Can help the manufactures understand the working capability of the machines and the track of their breakdown too.
- The unexpected breakdown of any machine can be easily prevented by keeping a track record of how and when the machines need maintenance and service so as to be able to function efficiently.

- **3. Quality Checks**

- can help in keeping a track of the quality checks before putting the product in the market and also can help in understanding the reasons or kinds of defects

10 Applications of Big Data in Manufacturing

• 4. Tool Optimization

- Big data tools equipped with adequate softwares and alerting systems can help in solving the root causes of the tools before they wear out too much and be added to the production cost for the repairs and problems.

• 5. Supply Chain Management

- will keep the suppliers in coordination and take the supplies as they are needed, this will prevent them from stockpiling and extra manufacturing which might do in vain and as well as is not so much of a sustainable practice.

• 6. Production Preparation

- Big data records can help them in forecasting the productions which might be needed in the market at a certain point of time so that they can produce the material beforehand and when the time is claimed they

10 Applications of Big Data in Manufacturing

- **7. Yield Improvement**

- Now, unlike production increase this has to do with the capacity increase without resources being increased. This goes on with keeping in mind the other factors that can help in production if the hidden patterns are understood clearly and widely. This will enable continuous improvement in the yield.

- **8. Market Study**

- Big Data analytics combined with IoT technological growth and other advanced analytics can help in understanding the market and how it is proving to be for this one industry.

- **9. Sustainable Development**

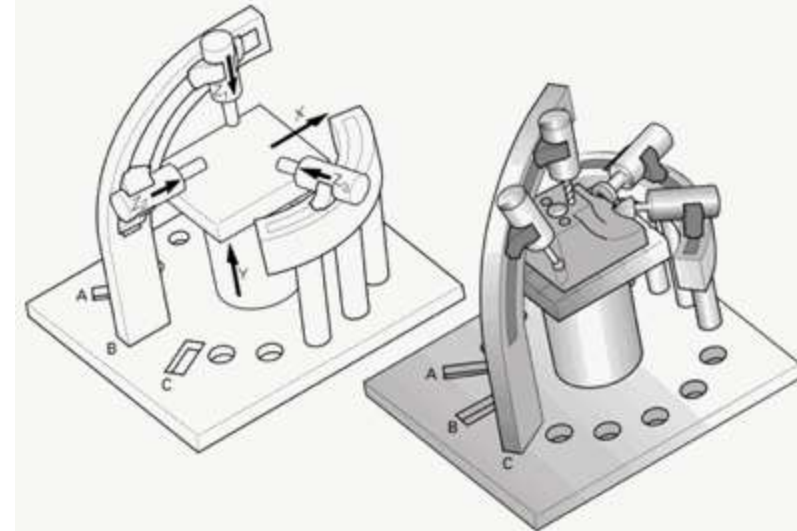
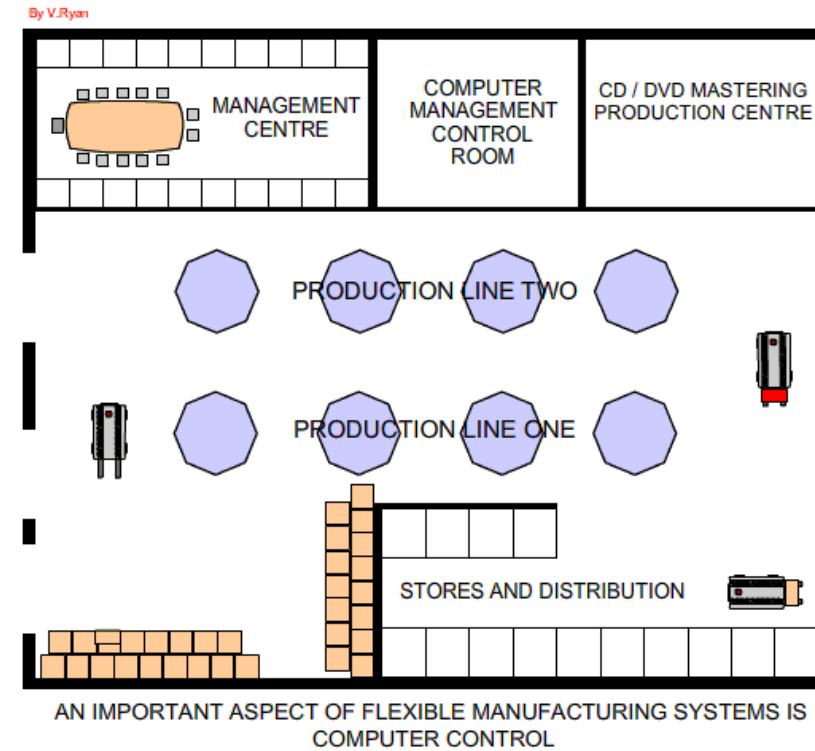
- Big data in the manufacturing industry if it gets deciphered properly can help in preventing the wastage of resources as the monitored activities will keep an eye on supply chain production management and even machine management to not be able to increase production cost and keep it the most efficient

- **10. Risk Evaluation**

- There are certain risks involved in the manufacturing industries be it in the form of casualties or in the form of

Technologies and their roles in smart manufacturing

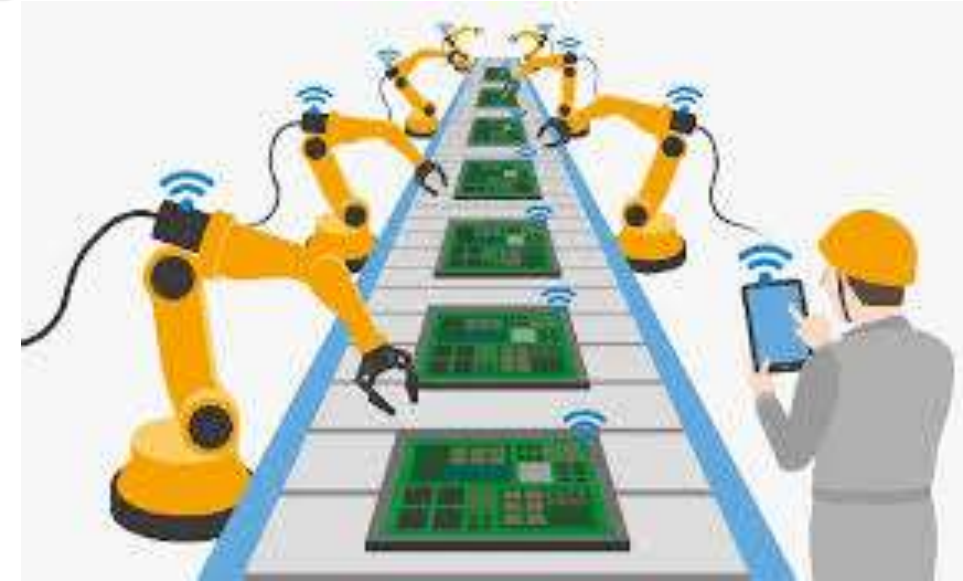
- 2.Flexible and Reconfigurable Manufacturing Systems (FRMS)
- Manufacturing system capable of adopting any variations occurred in built-in priority and functions due to the market demand changes or due to reconfiguration of product.
- They are focused on cost-effectiveness and quick response to rapid system changes.
- It is capable of production in small batch size and easily reconfigured for manufacture different variety of products which can be divided into two types;
- routine flexibility allows the manufacturing plant to produce new types of product in the same production line
- while machine flexibility allows to schedule the manufacturing routines of different machining stations, share the machining activity between them, replace the machining parts automatically.



Technologies and their roles in smart manufacturing

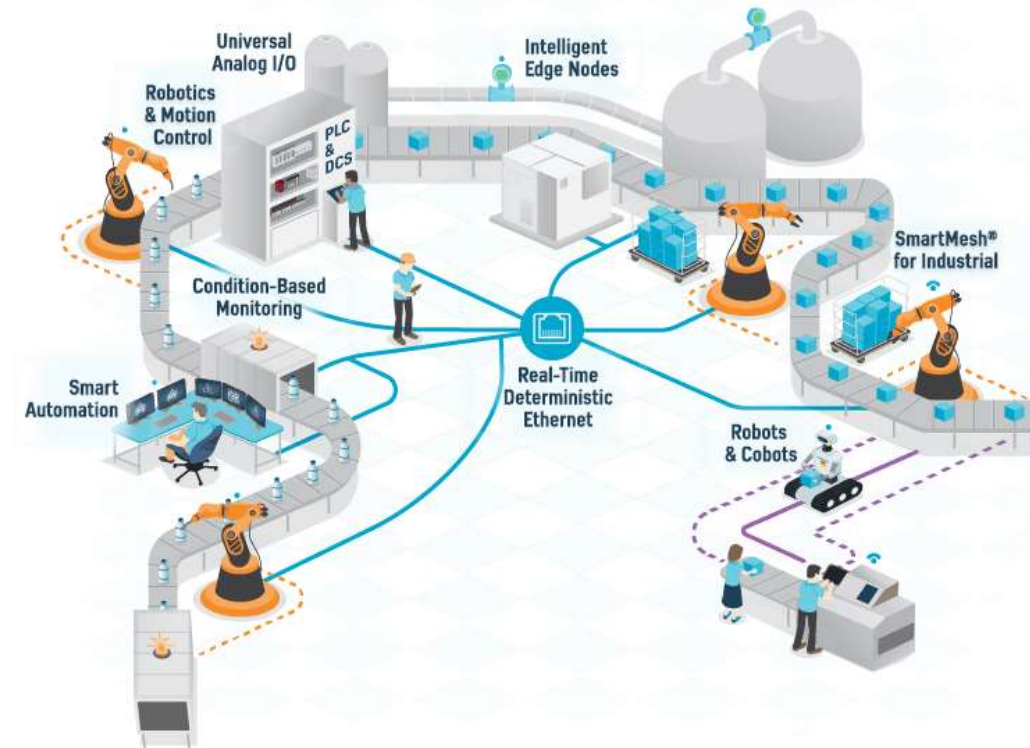
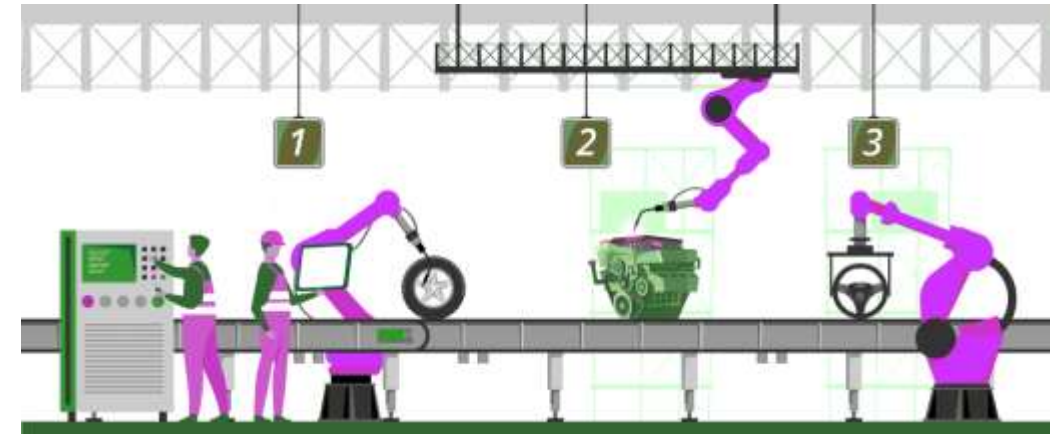
• 7. Artificial Intelligence

- For better human-robot collaboration and reduce the human workforce in risk zones, to improve the maintenance system of the manufacturing system and detection of any failures in the machinery and the product, artificial intelligence is adapted in new generation manufacturing system.
- The artificial intelligence system is capable of self-decision, self-optimizing and automatic response to physical changes such as changing the production schedules, stopping or running of any machining



Technologies and their roles in smart manufacturing

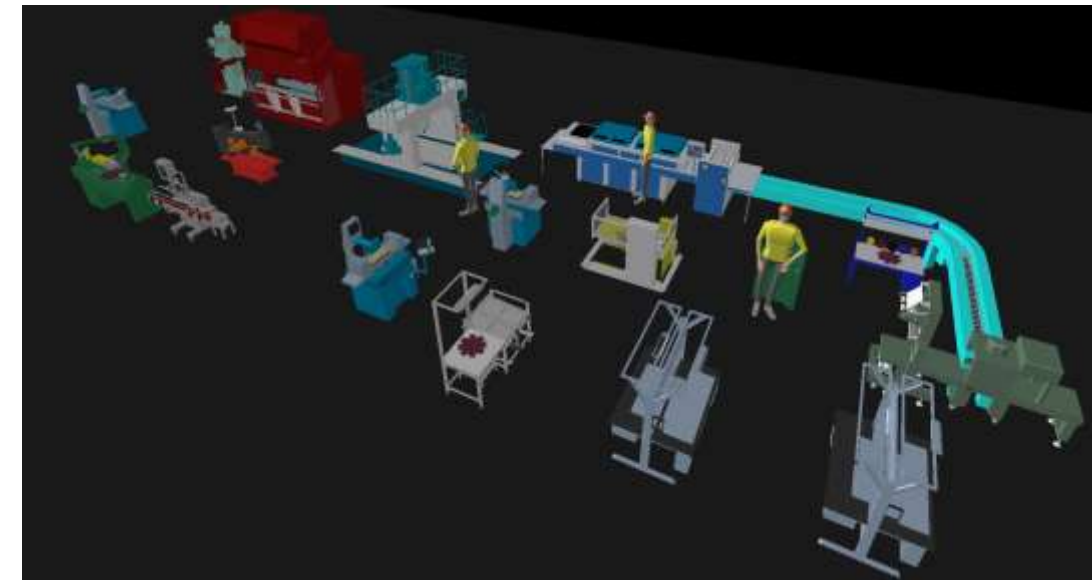
- 8.IoT and IIoT
- IoT is being used in common domestic applications like smart homes, transportation, logistics, healthcare, agriculture, human pets and vehicle tracking applications and
- IIoT is focused on the Industrial application of IoT connecting every physical entity with each other through the internet.
- In industries, the IIoT connects the physical entities like sensors, actuators and the entire process monitoring and control system into the internet cloud, enabling the interaction and cooperation of each entity together to reach the common goals.
- These interactions between each component help in production planning, predictive maintenance and faults localization, better human-machine interaction, intelligent process control for resource, tools and materials optimization.



Technologies and their roles in smart manufacturing

• 1. Simulation

- Simulation of plant operations to optimize the machine settings for preceding production line in a virtual environment without testing in physical world would save time and money in testing the production system.
- Simulations are used to mirror the physical world in a virtual model which can be used for production planning and scheduling according to obtained simulation results.
- Moreover, simulation has been implemented in SMS system these days which helped to analyze the design errors, production time, workforce and energy required for the entire production and facilitated the preparation of



APPLICATION OF **IOT** IN AUTOMOBILE INDUSTRY



Application of Internet of Things (IoT) in the Automotive Industry

- The automotive industry is changing a lot with usage of the Internet of Things (IoT).
- IoT is transforming how cars are designed, built, and used, making them safer, more efficient, and improving the overall driving experience.
- This shift is driving new innovations and creating new opportunities for both car makers and drivers.
- IoT has brought big changes to many industries, and the automotive sector is no different.
- It has improved how cars are made, operated, and maintained, adding better safety features, efficiency, and connectivity.
- In this blog, we'll look at 10 exciting ways IoT is used in the automotive industry. You'll see how these technologies are making cars smarter

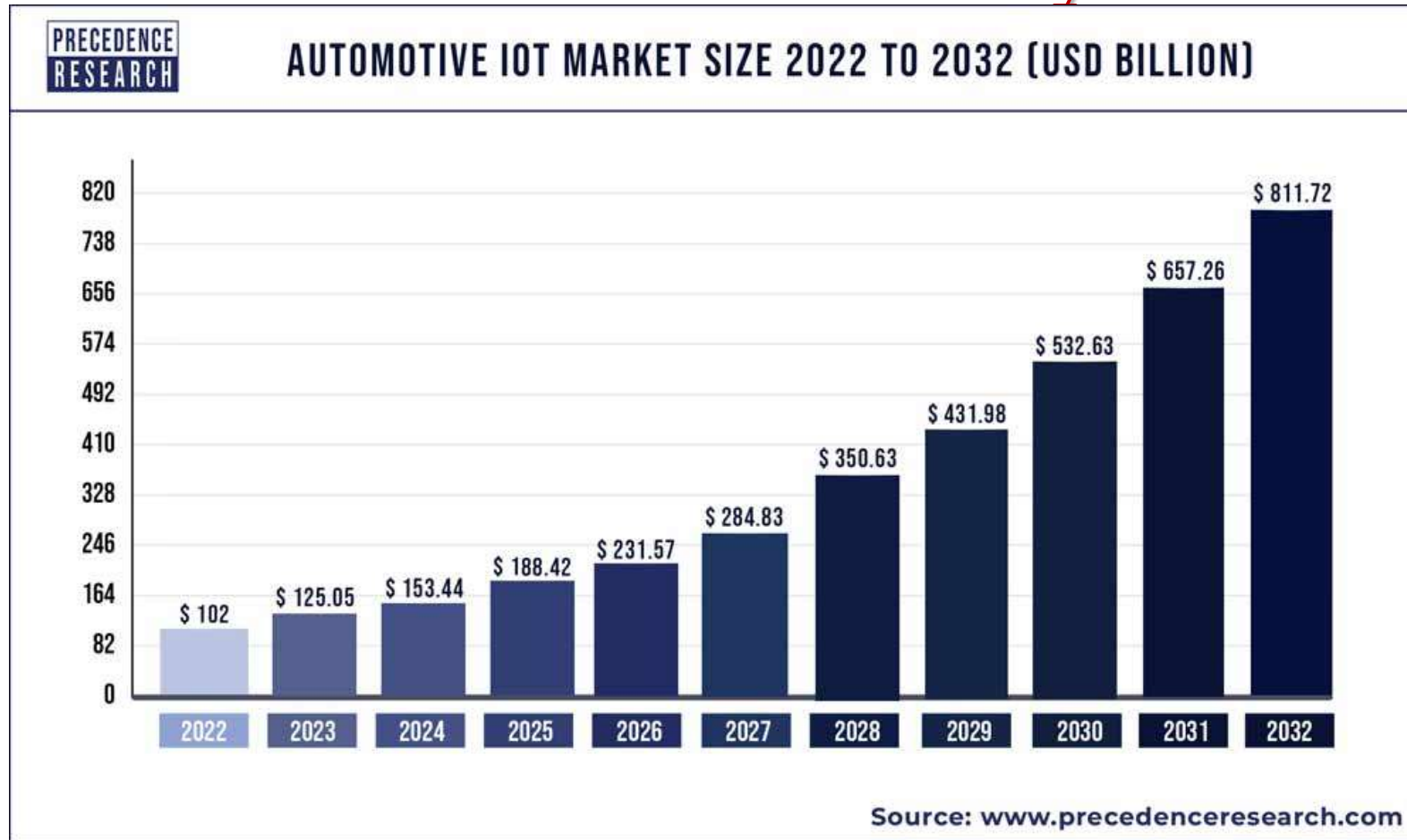
What is Automotive IoT?

- What is the Internet of Things?
- Internet of Things refers to a connection of billions of complex devices like electronics, sensors, gateways, actuators, and platform hubs.
- These tangible devices connect and interact with each other over a wireless network. Connected objects (or things) share data with each other and operate without any intervention by humans.
- Automotive IoT integrates devices like [sensors](#) and [GPS tracking](#) with software inside vehicles and across the automotive ecosystem.
- IoT applications use [embedded solutions](#) to deliver real-time data, optimizing the car manufacturing process and improving transport management.
- By adopting IoT, automobile manufacturers can automate production processes, utilize predictive maintenance to minimize downtime, and boost their bottom line.
- Similarly, [Automotive AI](#) benefits businesses such as fleet management companies, car rentals, suppliers, and

Application of Internet of Things (IoT) in the Automotive Industry

- Internet of Things is a method of obtaining data from several remote devices to automate and notify the user so he has a safety net protecting him from a potential failure.
- It is the emerging technology that can be found in every field, such as smartphone, household appliances, automated parking, commercial aviation, medical, farming, fertilizer, retail etc.
- We can develop a module that can be incorporated in automobile field to enhance the user experience of the vehicles, to assist the driver or to increase the comfort for the passengers, to safeguard the passengers or driver, to rescue the passengers in case of accidents, to provide secured parking, to avoid road rule violations, etc.

IoT In Automotive Industry Statistics



- The global automotive IoT market size was valued at USD 102 billion in 2022 and is projected to surpass around USD 811.72 billion by 2032.
- The global automotive IoT market is expected to grow at a compound annual growth rate (CAGR) of 23.10 during the forecast period 2023 to 2032.
- Major manufacturers of automobiles are concentrating on manufacturing vehicles that make use of software's. In order to obtain real time information which is extremely useful for the drivers there shall be an increase in the use of automotive IoT.

the automotive industry.

These technologies are making cars smarter and changing the way we interact with them.

- **1. Connectivity**

- IoT excels by allowing vehicles to connect smoothly with a range of technologies, including smartphones, home devices, other vehicles, and smart city infrastructure. This advanced connectivity is driven by the Vehicle-to-Everything (V2X) concept, which plays a crucial role in improving various aspects of driving.

- **2. Data Analytics**

- Data analytics in the automotive industry involves collecting and analyzing the extensive data generated by interconnected vehicles and their environments.

- **3. Fleet Management**

- IoT in fleet management systems offer a complete solution for businesses that rely on vehicle fleets. These systems provide real-time tracking, predictive maintenance, and driver behavior analysis, making them essential tools in the automotive industry's IoT applications.

- **4. Automation**

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- **5. Autonomous Vehicles**

- Autonomous or self-driving vehicles rely extensively on IoT to manage and utilize a massive stream of real-time data. IoT devices embedded in these vehicles continuously gather and transmit data from various sensors, including cameras, radar, and lidar.
- This data is crucial for several functions, such as real-time navigation, predictive modeling, and vehicle-to-vehicle communication.

- **6. Vehicle Telematics**

- One of the most significant use cases of IoT in automotive is the integration of cloud-connected IoT boxes or Telematics Control Units (TCUs) with telematics devices.
- This technology delivers real-time data on driver health, behavior, and vehicle status, which enhances efficiency, reduces costs, and improves safety.

- **7. In-vehicle Infotainment**

- In-vehicle infotainment is one of the most exciting IoT use cases in the automotive industry. It enhances the driving experience by providing entertainment, communication, and

is changing ways (use cases) IoT is used in the automotive industry.

These technologies are making cars smarter and changing the way we interact with them

• **8. Predictive Maintenance**

- IoT sensors actively monitor vehicle components in real time, constantly tracking key systems such as the engine, brakes, and fluid levels. By detecting early signs of potential issues like engine malfunctions or low oil, these sensors provide timely alerts before problems escalate.

• **9. Environmental Sustainability**

- IoT sensors and data analytics help the automotive industry reduce its environmental footprint. By monitoring emissions, fuel efficiency, and energy consumption, manufacturers can develop eco-friendly vehicles and cut their carbon footprint.

• **10. Safety and Security**

- IoT-based security systems track and monitor vehicles in real-time, preventing theft and improving recovery rates. They also enhance driver and passenger safety with advanced features like collision detection and automated emergency response

Conclusion

- The application of IoT in the automotive industry is transforming the way vehicles are designed, manufactured, and operated.
- These 10 use cases represent just a fraction of the possibilities that IoT offers to the automotive sector.
- As technology continues to advance, we can expect even more innovative solutions to emerge, further enhancing safety, efficiency, and the overall driving experience.
- The automotive industry's integration of IoT is a testament to its commitment to harnessing technology for a brighter, smarter future.
- In conclusion, the adoption of IoT in the automotive industry is not just a trend but a necessity to stay competitive and meet the evolving demands of both manufacturers and consumers.

Case Studies on Automotive Manufacturing and Industry 4.0

- Artificial intelligence (AI) and Industry 4.0 have been part of the automotive sector for years. Innovators combined these technologies with big data to cut costs through improved quality and production.
- Automotive manufacturers can transform their facilities into smart production plants with a top industrial IoT (IIoT) platform.
- Leading business digitalization platforms offer:
 - AI-powered visual inspection
 - Data orchestration
 - Enterprise integration
 - These platforms can locate the source of a

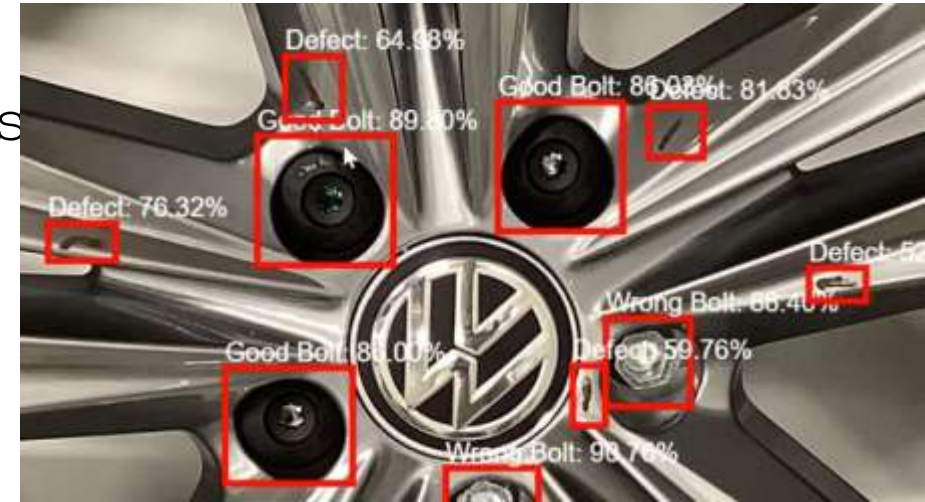
Automotive Manufacturing Case Study A: Lug Nuts

- The number of parts in a car can vary between 30,000 and 40,000, depending on the vehicle's make and model.
- The simplest defect in a part can have serious implications and delay production.
- Automotive manufacturers always look for solutions that minimize downtime and safeguard production schedules.
- One leading IoT platform includes a suite of AI applications – [a versatile software solution for visual inspection](#).
- This AI-powered application uses advanced algorithms and deep learning to find problems and solve issues in real time.
- Manufacturers train the inference engine.
- Powered by AI and machine learning (ML), it identifies defects humans often miss.

Automotive Manufacturing Case

Study A: Lug Nuts

- A common application of this visual inspection solution is checking assembled components.
- For example, inspecting wheels for missing or damaged lug nuts.
- Cameras take pictures of vehicles' wheels as they proceed along the production line.
- These photos go through the IoT platform and then to the inference engine.
- The inference engine analyzes the images and creates a markup that indicates the condition of the lug nuts.
- Edge communication sends the data to a system that generates a trouble ticket if needed.
- Best-in-class offerings provide configuration of customized dashboards and external communications with PLCs.



Automotive Manufacturing Case Study B: 24/7 Production Line Monitoring

- Automobile factories contain arrays of industrial equipment.
- Companies need real-time analytics to monitor their machines and the data they generate to avoid downtime and boost uptime.
- However, data is not useful unless connected to a business application.
- A metalworking and metal fabrication machine manufacturer has thousands of large, expensive machines in customers' factories worldwide.
- The manufacturer introduced a remote monitoring solution to supply customers with 24/7 production line monitoring.
- An IoT platform powers the solution.
- It includes edge software and a cloud-based



Automotive Manufacturing Case Study B: 24/7 Production Line Monitoring

- Customers' machines connect to the IoT provider's cloud via edge software.
- The cloud sends data to the machine tool manufacturer's application.
- Customers receive collected machine health and performance data so they can view their machines' real-time statuses.
- The solution features remote diagnosis and monitoring for quick intervention and on-site visits.
- Moreover, the IoT platform integrates seamlessly with the machine tool manufacturer's existing system's



CASE STUDY: The first fully digital automotive factory in Southeast Asia

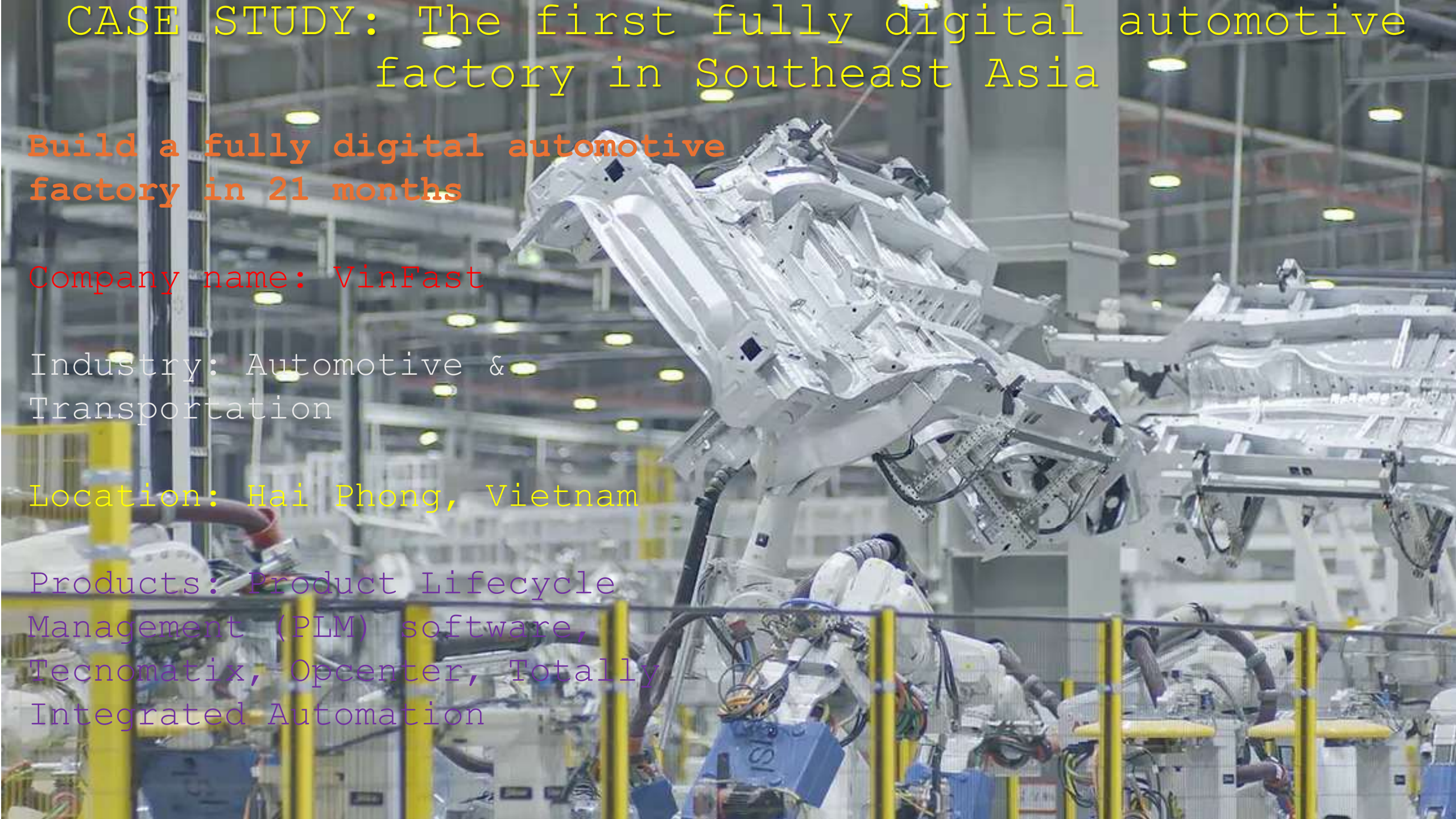
Build a fully digital automotive factory in 21 months

Company name: VinFast

Industry: Automotive & Transportation

Location: Hai Phong, Vietnam

Products: Product Lifecycle Management (PLM) software, Tecnomatix, Opcenter, Totally Integrated Automation



CASE STUDY: The first fully digital automotive factory in Southeast Asia

- As a member of Vingroup, one of Vietnam's largest privately held companies, VinFast teamed up with car manufacturers from America, Europe and Siemens to build their automotive factory in only 21 months.
- The holistic Siemens Xcelerator portfolio helped realizing the ambitious goal.
- **Prepared for regional and global competitiveness**
- Producing vehicles in a greenfield automotive factory in less than two years is unprecedented.
- However, VinFast wanted to be competitive in Vietnam and globally right from the beginning, so the company relied on Siemens' expertise to utilize the latest technology.
- This resulted in a closed-loop manufacturing system which uses digital twins of the products, the production, and the performance of production and product.
- The fully digital factory was built in 21 months, 50

CASE STUDY: The first fully digital automotive

factory in Southeast Asia

- Building a digital enterprise from scratch

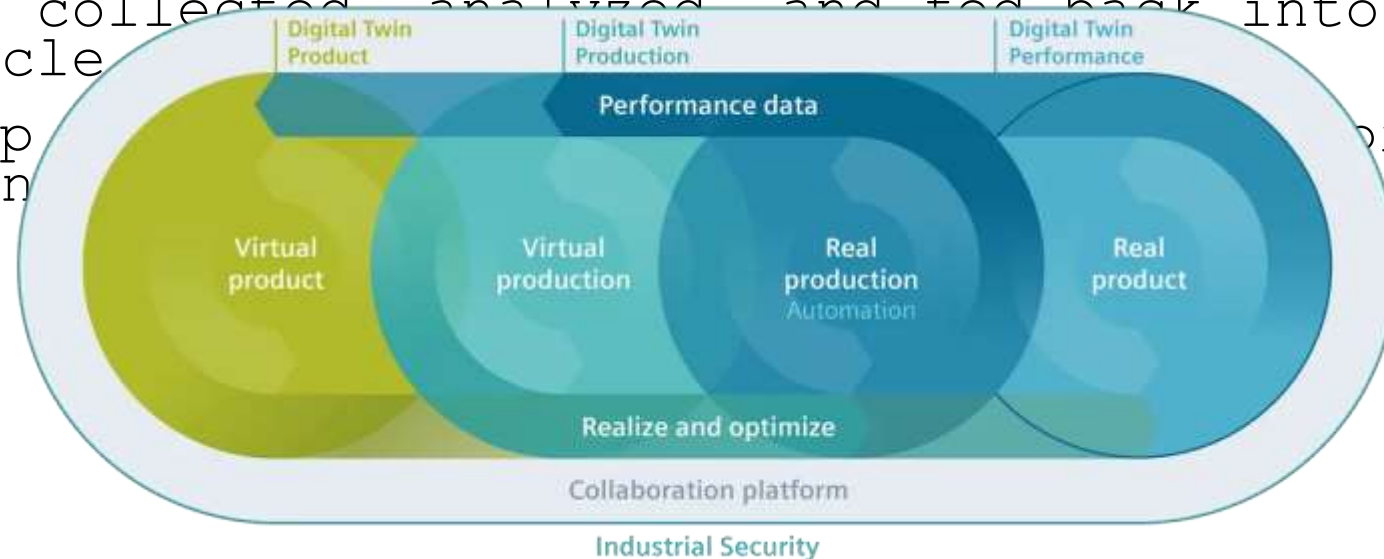
- Before VinFast's 335-hectare plant in Hai Phong, there was no Vietnamese brand for passenger cars.
- Siemens accepted the challenge to build a fully digital automotive factory within only 21 months.
- **Getting the most out of the Siemens Xcelerator portfolio**
- VinFast is using the comprehensive offerings from Siemens that combines Product Lifecycle Management (PLM) software such as the industry-leading Tecnomatix portfolio with Manufacturing Operations Management (MOM), through the new harmonized, holistic portfolio Siemens Opcenter, to realize lean manufacturing across all phases, and with Totally Integrated Automation for all automation, including robots, conveyors, presses and milling machines.
- This holistic approach has increased the speed and flexibility in development, ensured high global standards in production, optimized the manufacturing process, and

CASE STUDY: The first fully digital automotive factory in Southeast Asia



CASE STUDY: The first fully digital automotive factory in Southeast Asia

- Merging the virtual and the real world
- Developing new cars and scooters, planning the new plant, and finally producing with the help of digital tools creates a detailed virtual image, the digital twin.
- The digital twin creates new insights, with the combination of physics-based simulations with data analytics in a fully virtual environment.
- This makes it possible to realize innovations faster and more reliable, while also requiring significantly fewer real prototypes.
- Even more data are created when the product is being produced or a plant begins operation.
- These performance data of the real production and of the real product can be collected, analyzed, and fed back into the development cycle
- Here they help products and processes at an



CASE STUDY: The first fully digital automotive factory in Southeast Asia

- VinFast's way to the fast lane
- VinFast aims to become South East Asia's leading vehicle manufacturer and a remarkable new player in the global automotive industry.
- To realize their goal, VinFast put their trust into the Siemens Xcelerator portfolio.
- Realized with the Siemens Xcelerator portfolio from Siemens
- VinFast implemented Siemens for its manufacturing lines to paint shop, body shop, s shop.

